Preprint

LETTER

Gravity Field Dynamics: Complete Field Closure from Stellated Octahedron Topology

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Abstract

Every formulation of gravity contains two elements: the geometric structure of the gravitational field, and the requirement that the field must extend completely through space. We demonstrate that both follow from a single equation in combinatorics, $2(d+1) = 2^d$, whose only positive integer solution is $d = 3$. This solution fixes two structural constants: $K = 4$ vertices per simplex and $D = 3$ spatial dimensions. Binary encoding in three dimensions produces two interlocking tetrahedra (the stellated octahedron) whose face structure yields the $(3+1)$ spacetime decomposition and whose face normals are identically the tetrad of general relativity. The requirement that the field close through this geometry determines the coupling polynomial $f(k, s) = 1 + ks + k^2$, the gravitational constitutive law $\mu(x) = x/(x+1)$, and the complete covariant scalar-tensor action, all with zero free parameters. Expressed in bimetric language, the topological coupling fixes the five previously free Hassan–Rosen coefficients to $\{1, 0, 16, 0, 1\}$ (companion letter). Every numerical constant in the theory reduces to $K = 4$ and $D = 3$. Different coupling depths yield constitutive laws that differ by approximately 12% at the transition acceleration scale, measurable in galaxy rotation curves. The binary structure validated here was independently encoded in the I Ching trigram system (c. 2800–1000 BCE), predating formal binary arithmetic by millennia.

Introduction

Gravity Field Dynamics (GFD) is a covariant tetrad formalism that derives gravitational coupling laws from the discrete combinatorial geometry of the local tetrad frame. Like other tetrad formalisms, it operates with dual tetrahedral field chambers. What distinguishes it is that both the topology and the field closure condition are outputs, not inputs.

Every formulation of gravity, from Einstein's metric tensor to Hassan–Rosen bimetric theory, implicitly contains two elements:

1. A *topology*: the structure of the gravitational field (what the field IS).
2. A *field closure condition*: the requirement that the field must extend completely through space (what the field DOES).

In general relativity, the topology is the Riemannian manifold and the closure condition is that Einstein's equations hold at every spacetime point. In tetrad gravity, the topology is the vierbein and closure is the metric-compatible connection condition. In bimetric gravity, the topology is two coupled tetrads and closure is ghost-freedom; without it, the field cannot close consistently. In scalar-tensor theories, the topology is the scalar field configuration and closure is the constitutive law that governs the interpolation between gravitational regimes.

In each case, both the topology and the closure condition are supplied as starting points. They are inputs to the framework, not results derived from it.

This letter demonstrates that both follow from a single equation in combinatorics:

$$2(d+1) = 2^d. \tag{1}$$

The first four sections establish the topology (what the gravitational field IS): the uniqueness theorem forces $d = 3$, binary parity produces the stellated octahedron, face classification yields the $(3+1)$ spacetime decomposition, and the face normals are identically the tetrad of general relativity. The remaining sections establish what field closure determines: the coupling constants, the constitutive law, the complete covariant action, and a falsifiable prediction, all with zero free parameters.

A companion letter, *Gravity Field Dynamics and Bimetric Gravity: Dual Tetrahedral Field Chambers as the Key to the General Parameters* (Nelson 2026, submitted to *Classical and Quantum Gravity*) [1], provides the full coupling derivation and ghost-freedom proof. The present letter places those results within the larger chain from combinatorial identity to observational prediction. The computational companion `run_all_validations.py` independently verifies every claim.

1 The Uniqueness Theorem

Every formulation of gravity takes the number of spatial dimensions as a given. We begin by removing this input.

A simplex is the simplest possible shape in any dimension: a line segment in 1D, a triangle

in 2D, a tetrahedron in 3D. A hypercube is the generalization of a cube: a line segment in 1D, a square in 2D, a cube in 3D, a tesseract in 4D. In d dimensions, a simplex has $d+1$ vertices and a hypercube has 2^d vertices.

For two simplices to exactly fill a hypercube, their combined vertex count must equal the hypercube's vertex count. This counting condition is

$$2(d+1) = 2^d. \tag{2}$$

An exhaustive check (Table 1) shows that $d = 3$ is the unique positive integer solution. For $d \geq 4$, the exponential 2^d outruns the linear $2(d+1)$ and the two sides never meet again.

Table 1. Exhaustive check of the simplex-hypercube equation.

d	$2(d+1)$	2^d	Match?
1	4	2	No
2	6	4	No
3	8	8	**Yes**
4	10	16	No
5	12	32	No

From this unique solution, two structural constants follow immediately:

$$\begin{aligned} K &= d+1 = 4 \quad \text{(vertices per simplex)}, \\ D &= d = 3 \quad \text{(spatial dimensions)}. \end{aligned} \tag{3}$$

As a self-consistency check, the tetrahedron ($d = 3$ simplex) satisfies Euler's polyhedron formula: $V - E + F = 4 - 6 + 4 = 2$.

These two integers, $K = 4$ and $D = 3$, are the only structural parameters of the theory. No physics has been invoked; the argument is pure combinatorics. Everything that follows inherits from this equation.

2 Binary Topology of the Dual Tetrahedra

2.1 Parity partition

The uniqueness theorem fixes $d = 3$ and $K = 4$, but says nothing about geometry. The question is what structure binary arithmetic in three dimensions produces.

The hypercube in $d = 3$ is the ordinary cube, with $2^3 = 8$ vertices. Each vertex can be addressed by a 3-bit binary string (000 through 111), where the three bits encode position along the x, y, and z axes: 0 for -1 and 1 for $+1$. Table 2 lists all eight vertices with their binary addresses and Cartesian coordinates.

Count the number of 1-bits in each vertex's binary address. If the count is odd (vertices 1, 2, 4, 7), the vertex belongs to one group; if even (vertices 0, 3, 5, 6), the other. This is the popcount parity partition.

Within each group, every pair of vertices differs in exactly 2 out of 3 bit positions (Hamming distance 2). In Cartesian coordinates, this uniformity guarantees that the four vertices of each group form a perfectly regular tetrahedron with edge length $2\sqrt{2}$. The two groups are vertex-disjoint, edge-disjoint, and share no faces; embedded in the cube, they form the stellated octahedron (stella octangula). This geometry was not designed; it was inherited from binary arithmetic in $d = 3$.

Table 2. The eight cube vertices. Binary address, coordinates, parity (number of 1-bits), and tetrahedron membership.

v	Binary	(x, y, z)	Parity	Tet
0	000	$(-1, -1, -1)$	even	B
1	001	$(-1, -1, +1)$	odd	A
2	010	$(-1, +1, -1)$	odd	A
3	011	$(-1, +1, +1)$	even	B
4	100	$(+1, -1, -1)$	odd	A
5	101	$(+1, -1, +1)$	even	B
6	110	$(+1, +1, -1)$	even	B
7	111	$(+1, +1, +1)$	odd	A

2.2 Structural constants

Every corner of the cube has a partner diagonally opposite. In binary: 000 pairs with 111, 001 with 110, 010 with 101, 011 with 100. Each pair sums to $7 = 111_2$. These four body diagonals all pass through the exact center of the cube, the point equidistant from all eight corners; all eight vertex position vectors cancel there. This point is the Field Origin, the unique location of maximal symmetry where the two chambers couple.

Each tetrahedron has $\binom{4}{2} = 6$ edges (every corner connected to every other corner). Two disjoint tetrahedra contribute 12 edges total, all of uniform length $2\sqrt{2}$. The complete topological inventory is

$$12 \text{ edges} + 1 \text{ origin} = 13 = KD + 1. \tag{4}$$

Every structural constant of the stellated octahedron reduces to K and D: $K = 4$ faces per tetrahedron, $2^D = 8$ vertices, $KD = 12$ edges, and $KD + 1 = 13$ structural elements. This number reappears as the coupling denominator.

3 Emergence of (3+1) Spacetime

The $(3 + 1)$ decomposition of spacetime is standard across general relativity, tetrad gravity, and bimetric gravity. It is also an output of the stellated octahedron: the decomposition follows from how the faces are oriented relative to their shared center.

A tetrahedron has four triangular faces, one opposite each of its four vertices. The eight total faces (four per tetrahedron) are the surfaces through which the two chambers interact. Of the four faces of each tetrahedron, three face outward (away from the center of the cube) and

one faces inward (toward the center, where the two tetrahedra meet). This inward-facing face is geometrically special: it is the surface that directly confronts the partner tetrahedron.

For TetA, the inward face is opposite vertex V_7 (the apex, at corner 111). For TetB, the inward face is opposite vertex V_0 (corner 000). These two inward faces are the time faces; the remaining six are spatial faces. Per tetrahedron, the count is

$$1 \text{ time} + 3 \text{ spatial} = (3+1). \tag{5}$$

The time fraction is $2/8 = 1/4$. Applied to the total orbital angular momentum quantum $2\hbar$, this yields spin $= (1/4) \times 2\hbar = \hbar/2$. The $(3+1)$ spacetime signature and the spin-1/2 value are topological consequences of the face classification, not additional inputs.

4 The Tetrad Connection

4.1 Coordinate representation

At every point in curved spacetime, one can erect a local scaffolding of four perpendicular directions, like four rulers meeting at a corner. This scaffolding is the tetrad ($e^a{}_\mu$, also called the vierbein). Einstein's metric, the rule that defines distances, is built from it:

$$g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu. \tag{6}$$

The tetrad is more fundamental than the metric because matter particles (electrons, quarks) can couple to gravity only through it. The tetrad formulation has been standard since the work of Einstein [3] and Weyl [4].

In the local orthonormal frame, the spatial components of a regular tetrad point along four directions: the four face normals of a tetrahedron. These directions can be written as four linear projections of the local coordinates:

$$\begin{aligned} v_1 &= x + y + z, \\ v_2 &= x - y - z, \\ v_3 &= -x + y - z, \\ v_4 &= -x - y + z. \end{aligned} \tag{7}$$

The primary field chamber (TetA) is the region where all four projections are bounded:

$$\text{TetA}: \quad \max(v_1, v_2, v_3, v_4) \le 1. \tag{8}$$

The inverted dual (TetB) satisfies the opposite inequality:

$$\text{TetB}: \quad \max(-v_1, -v_2, -v_3, -v_4) \le 1. \tag{9}$$

Their intersection is the central octahedral region (the Field Origin). Their union is the stellated octahedron.

Figure 1 shows the result of evaluating these half-space inequalities at four million randomly sampled points. No mesh is constructed; the geometry emerges from the boundaries alone. On each face of TetA, the penetration by TetB's opposing vertex subdivides the surface into four sub-regions: three spatial corners and one central inverted core. This $3+1$ subdivision, repeated across all four faces, produces $4 \times 4 = 16$ independent boundary channels.

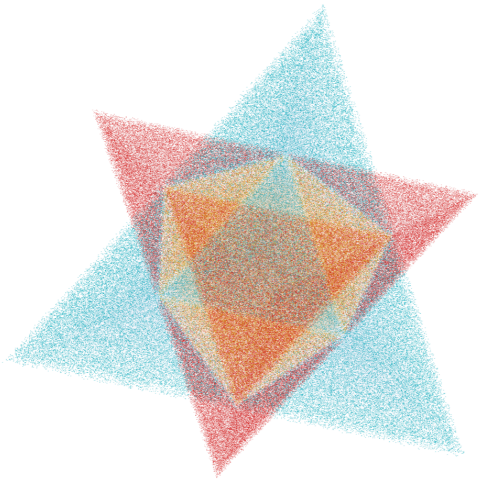


Figure 1. The dual tetrahedral field chamber. Four million points are sampled uniformly in local frame coordinates and tested against the half-space inequalities $L_i(\mathbf{p})$ defined in the text. Points that satisfy the TetA bounds are shown in cyan; those satisfying TetB in red; and those inside both (the shared octahedral Field Origin) in gold. No mesh or curve fitting is used. Each face of TetA is subdivided into three exposed corners and one inverted core where the opposing vertex penetrates, giving $4 \times 4 = 16$ geometric channels. These channels determine the gravitational coupling structure; expressed in bimetric language, they fix the interaction potential. An interactive reconstruction is provided as supplementary material.

4.2 Face normals and binary vertices

The four face-normal directions (v_1 through v_4) each contain three signs (+ or −). Reading + as 1 and − as 0 converts each normal into a 3-bit binary address. Table 3 shows the complete mapping.

Table 3. Sign-to-binary encoding of the tetrad face normals.

Normal	Signs	Binary	Decimal	Tet
$v_1 = +x+y+z$	+ + +	111	7	A
$v_2 = +x-y-z$	+ − −	100	4	A
$v_3 = -x+y-z$	− + −	010	2	A
$v_4 = -x-y+z$	− − +	001	1	A
$-v_1$	− − −	000	0	B
$-v_2$	− + +	011	3	B
$-v_3$	+ − +	101	5	B
$-v_4$	+ + −	110	6	B

Each face of the GR tetrad points toward one specific corner of the cube. The four face

normals of the primary tetrad point to corners 7, 4, 2, 1: the four odd-parity corners, TetA $= \{7, 4, 2, 1\}$. Negating all normals yields corners 0, 3, 5, 6: the four even-parity corners, TetB $= \{0, 3, 5, 6\}$. The stellated octahedron derived in Sections 1–3 from binary arithmetic and the tetrad pair that general relativity has used since 1928 are the same mathematical object.

We have established what the gravitational field IS: the stellated octahedron, forced by the uniqueness theorem and identical to the GR tetrad. But knowing what a field IS does not tell us what it DOES. A gravitational field created by a mass does not stop at an arbitrary boundary; it extends through all of space, accounting for every point. This completeness requirement is *field closure*. In Einstein's general relativity, field closure is enforced by requiring the field equations to hold at every spacetime point. We now ask a different question: what does the closure requirement *determine*?

5 Coupling Constants from Geometry

5.1 The locked parameter set

For the field to close, the two chambers must couple. The geometry of the dual tetrahedral field chamber determines how.

TetB's penetrating vertex subdivides each face of TetA. Take any triangle and connect the midpoints of its three edges; this creates four smaller triangles, all identical in shape to the original (the medial triangle theorem). When TetB's vertex penetrates a face of TetA, it sits at the center of this four-part subdivision. Each of TetA's $K = 4$ faces acquires 4 sub-channels, for a total of

$$K \times K = K^2 = 16 \tag{10}$$

independent coupling channels at the face-to-face boundary.

The two interlocking tetrahedra correspond, in the language of bimetric gravity [2], to two interacting metrics with an interaction potential governed by five coefficients β_0 through β_4. These have been free parameters for over a decade. The closure geometry determines all five:

- $\beta_0 = \beta_4 = 1$: exchange symmetry between the two chambers fixes the boundary terms at unity.
- $\beta_1 = \beta_3 = 0$: the two tetrahedra have opposite spatial orientations; orientation parity at the Field Origin forces all odd-order mixing to vanish.
- $\beta_2 = K^2 = 16$: the pairwise face-to-face coupling absorbs the full channel count.

The complete locked set is

$$(\beta_0,\, \beta_1,\, \beta_2,\, \beta_3,\, \beta_4) = (1,\, 0,\, 16,\, 0,\, 1). \tag{11}$$

The detailed derivation and ghost-freedom proof appear in the companion letter [1]. A ghost is a theoretical excitation with negative energy; if present, the vacuum is unstable and the field cannot close consistently [10]. Ghost-freedom ensures the field CAN close; it also guarantees the existence of a Vainshtein screening mechanism [9] that recovers general relativity in strong-field environments. The locked potential lies within the ghost-free Hassan–Rosen class by construction.

5.2 The coupling polynomial

The coupling polynomial counts the number of active geometric channels between the two tetrahedra as a function of the field's internal state. When the field is at full strength ($s = +1$), all channels are active. At the symmetric center ($s = 0$), only the even-order channels contribute. At the lower bound ($s = -1$), the count reaches its minimum. The polynomial encoding this spectrum is

$$f(k,\, s) = 1 + ks + k^2. \tag{12}$$

At $k = K = 4$, it evaluates to three structural states:

$$\begin{aligned} f(4,\, +1) &= 21 \quad \text{(full capacity)}, \\ f(4,\, 0) &= 17 \quad \text{(Field Origin)}, \\ f(4,\, -1) &= 13 \quad \text{(geometric minimum)}. \end{aligned} \tag{13}$$

The closure range is $f(+1) - f(-1) = 21 - 13 = 8 = 2K = 2^D$: both twice the closure constant and the total vertex count. The field must traverse this full range to close.

The ratio of the two central values carries an algebraic signature. A continued fraction is a way of writing a number as nested divisions; for example, $17/13 = 1 + 1/(3 + 1/4)$, written $[1;\, 3,\, 4]$. The partial quotients 1, 3, 4 encode the number's internal structure. For the gravitational ratio 17/13, these quotients are

$$\frac{17}{13} = [1;\, D,\, K] = [1;\, 3,\, 4], \tag{14}$$

literally encoding both structural parameters from the uniqueness theorem.

5.3 Algebraic structure of the coupling polynomial

The coupling polynomial $f(k, s) = 1 + ks + k^2$ has a property that places it in a well-studied family. Reverse the order of its coefficients: $k^2 + sk + 1$ has the same coefficients read forward and backward. A polynomial with this

symmetry is called *self-reciprocal*: it satisfies $k^2 f(1/k, s) = f(k, s)$. Self-reciprocal polynomials are standard objects in algebra; their defining consequence is that if r is a root, so is $1/r$.

Treating f as a quadratic in k, the roots are $k = (-s \pm \sqrt{s^2 - 4})/2$. For every physical state $|s| \leq 1$, the discriminant $s^2 - 4$ is negative, so the roots are complex conjugates. Combined with self-reciprocity, $|r|^2 = r \cdot (1/r) = 1$: every root lies on the unit circle in the complex plane, for all s in $[-1, +1]$.

At the two capacity endpoints, the polynomial reduces to classical cyclotomic polynomials — the polynomials whose roots are the primitive nth roots of unity:

$$\begin{aligned} f(k, +1) &= k^2 + k + 1 = \Phi_3(k), \\ f(k, -1) &= k^2 - k + 1 = \Phi_6(k). \end{aligned} \tag{15}$$

Φ_3 is the third cyclotomic polynomial; its roots are the primitive cube roots of unity. Φ_6 is the sixth; its roots are the primitive sixth roots. These are textbook objects, tabulated in any algebra reference. At the capacity endpoints, the root phases are $\pi/3$ and $2\pi/3$ — two adjacent vertices of the regular hexagon inscribed in the unit circle. At the Field Origin ($s = 0$), the roots sit at $\pm i$ (phase $\pi/2$), the midpoint of the arc between them. As s sweeps from -1 to $+1$, the roots trace exactly one hexagonal sector (60 degrees) of the unit circle.

The product identity $\Phi_3(k) \cdot \Phi_6(k) = k^4 + k^2 + 1$ links the two capacity endpoints algebraically. At $k = K = 4$, this reads $21 \times 13 = 273 = 4^4 + 4^2 + 1$, confirming that the full capacity and the geometric minimum are not independent numbers but two evaluations of a single cyclotomic structure.

The coupling polynomial belongs to the cyclotomic family — the family that encodes rotational symmetry. This is a consequence of the stellated octahedron's geometry, not a design choice.

6 The Scalar-Tensor Action

The coupling polynomial is a static encoding. The field must close dynamically. To extract the dynamics, we project the closure parameter onto observable quantities.

In materials science, a constitutive law relates stress to strain: it tells you how a material responds to force. In gravity, the constitutive law $\mu(x)$ plays the same role: it specifies what fraction of the expected Newtonian acceleration is actually produced at a given field strength. The dimensionless kinematic acceleration $x = |g|/a_0$ measures the local gravitational field in units of the transition acceleration scale ($a_0 \approx 1.2 \times 10^{-10}$ m/s^2). Near a massive object ($x \gg 1$), μ approaches 1 and gravity is fully Newtonian. Far from any source ($x \ll 1$), μ drops below 1 and gravity departs from Newton's predictions. This interpolation structure was first proposed by Milgrom [5]; Bekenstein and Milgrom [6] showed it could be derived from a scalar-tensor action.

6.1 *From topology to the constitutive law and covariant action*

The observable quantity x ranges from zero to infinity. The internal state parameter s ranges from -1 to $+1$. The unique smooth map connecting these two ranges while fixing the transition point ($x = 1$, $s = 0$) at the center is the Mobius transformation

$$s(x) = \frac{x - 1}{x + 1}. \tag{16}$$

Composing $f(K, s(x))$ gives the continuous topological capacity:

$$f(x) = \frac{21x + 13}{x + 1}, \tag{17}$$

which interpolates smoothly between the geometric minimum ($f = 13$ at $x = 0$) and the full capacity ($f \to 21$ as $x \to \infty$).

Normalizing $f(x)$ to the range $[0, 1]$ asks: what fraction of the total capacity range is being used? The formula is $\mu = (f - 13)/(21 - 13) = (f - 13)/8$. Substituting:

$$\begin{aligned} \mu(x) &= \frac{1}{8}\left(\frac{21x + 13}{x + 1} - 13\right) \\ &= \frac{1}{8} \cdot \frac{21x + 13 - 13x - 13}{x + 1} \\ &= \frac{1}{8} \cdot \frac{8x}{x + 1} = \frac{x}{x + 1}. \end{aligned} \tag{18}$$

The factor of 8 cancels perfectly. This cancellation is not a coincidence; it is the vertex count 2^D eliminating itself from the constitutive law, leaving a function determined entirely by the topology.

The constitutive law tells us the local field response. A complete gravitational theory also requires an action, a Lagrangian from which the field equations follow variationally. Integrating the constitutive law via $\mu(\sqrt{y}) = dF/dy$ (with $y = |\nabla\Phi|^2/a_0^2$) yields the kinetic functional

$$F(y) = y - 2\sqrt{y} + 2\ln(1 + \sqrt{y}), \tag{19}$$

with the boundary condition $F(0) = 0$ fixing the integration constant. The complete covariant scalar-tensor action is then

$$S = \int d^4x \sqrt{-g}\left[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G} F\left(\frac{|\nabla\Phi|^2}{a_0^2}\right) + \mathcal{L}_{\mathrm{m}}\right]. \tag{20}$$

The three coupling scales in this action are $16\pi G$, $8\pi G$, and $1/4$, with numerical prefactors 16, 8, 4. These are K^2, $2K$, K: the face-channel count, the vertex count, and the face count. Each is half the previous. The hierarchy of gravitational coupling strengths is the counting sequence of the stellated octahedron.

Every coefficient in the action is fixed. The constitutive law governs local response; the action governs global dynamics. Neither is fitted; both follow from the stellated octahedron's closure capacity.

7 Structural Completeness

Claiming zero free parameters requires proof. Table 4 exhibits every numerical constant in the theory as a function of $K = 4$ and $D = 3$.

Table 4. Every constant in the theory reduces to K and D.

Value	Expression	Physical meaning
4	K	Faces per tetrahedron
8	2^D	Vertices of the cube
12	KD	Edges of the stellated octahedron
13	$KD + 1$	Edges + Field Origin
16	K^2	Coupling channels
17	$K^2 + 1$	Capacity at Field Origin
21	$K^2 + K + 1$	Full capacity
64	K^D	Vertex-pair state space

The three gravitational coupling scales (16, 8, 4) form the geometric halving sequence K^2, $2K$, K: face channels, vertices, faces. The continued fraction $17/13 = [1; D, K]$ encodes both structural parameters as partial quotients. The theory has exactly two structural parameters, both fixed by the uniqueness theorem. Closure is total: no constant is imported from outside the stellated octahedron.

8 Observational Predictions

A theory determined by topology and closure alone must still submit to experiment. The N-metric screening family provides the testing ground.

Stars at the edges of galaxies orbit faster than Newtonian gravity predicts from visible matter alone. A rotation curve plots orbital speed versus distance from the galactic center; Newtonian theory predicts a decline at large distances, but observed curves remain flat. This discrepancy is the primary observational test of gravity at galactic scales.

Near the Sun, gravity behaves exactly as Newton described. At the outskirts of a galaxy, it does not. The transition between these regimes is called screening: the modified gravitational effects are screened (hidden) in strong-field environments. The screening equation determines where this transition occurs.

The generalized screening equation

$$x(1-\mu)^N = \mu \tag{21}$$

arises when N dynamical metric sectors participate in the closure. At $N = 1$, the theory reduces to a scalar-tensor formulation (one metric responds to the scalar field), recovering $\mu(x) = x/(x+1)$. At $N = 2$, both metrics are dynamical, recovering the full bimetric theory. Different closure depths yield different constitutive laws.

At the transition scale $x = 1$, the $N = 1$ theory gives $\mu(1) = 1/2$. For $N = 2$, the screening equation becomes $\mu^2 - 3\mu + 1 = 0$, a quadratic whose physical root is

$$\mu(1) = \frac{3-\sqrt{5}}{2} = \frac{1}{\varphi^2} \approx 0.382, \tag{22}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

Away from the transition point, the $N = 2$ screening equation can be solved in closed form. The full constitutive law for the bimetric theory is

$$\mu_2(x) = \frac{\sqrt{1+4x}-1}{\sqrt{1+4x}+1}, \tag{23}$$

which satisfies $x(1-\mu_2)^2 = \mu_2$ identically. At $x = 1$, this reduces to $(\sqrt{5}-1)/(\sqrt{5}+1) = 1/\varphi^2$, recovering the golden ratio root above.

Both constitutive laws share the same asymptotic limits. In the Newtonian regime ($x \gg 1$), both approach $\mu \to 1$: gravity is unmodified near massive objects, consistent with solar system tests. In the deep-field regime ($x \ll 1$), both approach $\mu \to x$: the linear scaling that produces flat rotation curves. The two theories are indistinguishable in both extremes. They differ at the transition scale, where $x \approx 1$ and the gravitational field is changing character.

At the transition acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2, the $N = 1$ theory predicts that gravity is at 50% of Newtonian strength, while $N = 2$ predicts 38.2%: a separation of 11.8% in absolute terms. This difference falls in the regime where the SPARC catalog [8] of 175 galaxies provides the densest coverage, making it a concrete target for observational discrimination.

9 Pi and Field Closure

The topology is the track. The closure requirement demands the circuit be completed. Pi provides the mechanism.

Just as numbers are normally written in base 10 (digits 0–9), they can be written in base 8 (digits 0–7). The stellated octahedron has 8 vertices, labeled 0 through 7. Pi in base 8

(3.11037…) produces a sequence of octal digits, each naming one of the eight vertices.

Starting at vertex 0, the first octal digit of pi after the decimal point is 1: move to vertex $(0+1) \bmod 8 = 1$. The next digit is 1: move to $(1+1) \bmod 8 = 2$. The next is 0: stay at $(2+0) \bmod 8 = 2$. The next is 3: move to $(2+3) \bmod 8 = 5$. Continuing this cumulative-sum-mod-8 walk, every vertex is visited within finitely many steps. Over many digits, each of the 8 vertices is visited with approximately equal frequency; no vertex is favored or avoided.

Under this traversal, the time face $\{1, 2, 4\}$ completes first (at step 5). Temporal structure is established before full spatial closure, consistent with the $(3+1)$ face classification of Section 3.

9.1 The BBP bridge

In 1997, Bailey, Borwein, and Plouffe [7] discovered a formula that computes any individual hexadecimal (base-16) digit of pi without first computing all the preceding digits. This formula was a result in pure number theory, with no connection to geometry or gravity. Its internal structural constants are:

- Period: $8 = 2^D$ (the vertex count).
- Active terms per period: $4 = K$ (the face count).
- Geometric series base: $16 = K^2 = 2^K$ (the channel count).
- Denominator coefficients: $\{+4, -2, -1, -1\}$, which sum to zero (Field Origin symmetry) and have magnitudes $\{4, 2, 1\}$, the binary place values that define the 3-bit coordinate encoding.

The three structural constants of the BBP formula (8, 4, 16) are exactly the vertex count, face count, and channel count of the stellated octahedron. The two frames (base-8 traversal and base-16 extraction) realign every $\mathrm{lcm}(3, 4) = 12 = KD$ binary digits: the edge count of the stellated octahedron.

The BBP formula was discovered independently of any geometric context. The coincidence of its structural constants with those of the stellated octahedron constitutes independent analytical confirmation.

10 Discussion

10.1 Summary and outlook

The equation $2(d+1) = 2^d$ has a unique solution $d = 3$, which forces the stellated octahedron, which is the GR tetrad, whose face geometry locks the coupling constants, whose normalized capacity is the constitutive law, whose integration yields the covariant action, and whose N-metric embedding yields a falsifiable prediction. The chain is:

$$d{=}3 \;\to\; \text{stellated oct.} \;\to\; (3{+}1) \;\to\; \text{tetrad} \;\to\; \{\beta_n\} \;\to\; \Phi_3/\Phi_6 \;\to\; \mu(x) \;\to\; S_{\mathrm{SST}}$$

Each link is a theorem or a necessary consequence:

1. The uniqueness theorem fixes $d = 3$ (Section 1).
2. Binary parity in $d = 3$ produces the stellated octahedron (Section 2).
3. Face classification yields the $(3+1)$ spacetime decomposition (Section 3).
4. The face normals are identically the GR tetrad (Section 4).
5. Face geometry locks all five bimetric coefficients $\{\beta_n\} = \{1, 0, 16, 0, 1\}$ (Section 5).
6. The coupling polynomial is cyclotomic (Φ_3 and Φ_6), with roots on the unit circle (Section 5).
7. Capacity normalization yields $\mu(x) = x/(x+1)$ (Section 6).
8. Integration gives the kinetic functional and the covariant action (20) (Section 6).
9. The N-metric family produces a falsifiable 11.8% separation at the transition scale (Section 8).

The reader can verify each step independently using the companion script `run_all_validations.py`, which tests all nine links in sequence.

Every gravity formalism contains two elements: a topology and a closure condition. Table 5 lists five formalisms with their topology and closure inputs. Gravity Field Dynamics is the first in which both are derived.

Table 5. Topology and field closure across gravity formalisms.

Formalism	Topology	Closure
GR	Riem. manifold	Field eqs. everywhere
Tetrad	Vierbein	Metric compat.
Bimetric	Two tetrads	Ghost-freedom
Scalar-tensor	Scalar field	Constitutive law
GFD	Stell. oct. *(der.)*	Cap. norm. *(der.)*

Three open questions remain: experimental discrimination at the transition scale via SPARC data; the extension to quantum gravity (where the closure frequency $\sim 10^{20}$ Hz becomes relevant); and the explicit closed-form EFT matching from the locked Galileon hierarchy to the IR constitutive law, framed as a checkable matching problem in the companion letter [1].

10.2 Historical note: the I Ching state space

The I Ching (c. 2800–1000 BCE) assigns eight trigrams to 3-line binary symbols. Each trigram is a stack of three lines, broken (0) or solid (1). Reading bottom-to-top as LSB-to-MSB, the eight trigrams map bijectively to the eight cube vertices 0–7. Table 6 shows the correspondence.

Table 6. I Ching trigrams mapped to cube vertices.

Trigram	Binary	Decimal	Name	Tet
☰	000	0	Earth	B
☶	001	1	Mountain	A
☵	010	2	Water	A
☴	011	3	Wind	B
☳	100	4	Thunder	A
☲	101	5	Fire	B
☱	110	6	Lake	B
☷	111	7	Heaven	A

The traditional Yin/Yang classification (even vs. odd line-count) is exactly the popcount parity partition that produces the dual tetrahedra.

Each hexagram in the I Ching consists of two stacked trigrams — a lower trigram and an upper trigram. In the vertex language developed here, this is an ordered pair of vertices: a state transition from one point in the cube to another. Since each trigram can be any of the eight vertices, there are $8 \times 8 = 64$ ordered transitions, and the I Ching enumerates all of them. The number admits three equivalent decompositions: 8^2 (all ordered vertex pairs), 2^6 (six-bit binary states), and $K^D = 4^3$ (the coupling constant raised to the spatial dimension).

These 64 transitions partition into four blocks of $K^2 = 16$ by chamber membership: TetA → TetA, TetA → TetB, TetB → TetA, and TetB → TetB. The 32 cross-chamber transitions represent couplings across the Field Origin; the 32 intra-chamber transitions represent reconfigurations within a single tetrahedron. When every one of the 64 has been realized, the system has completed one full closure cycle through the entire state space. The companion validation script confirms that the π base-8 traversal (Section 9) generates all 64 transitions, with balanced frequency across both chambers.

The system captures the full $4^3 = 64$ transition space of a three-dimensional binary field — dual-chamber partition, parity classification, and complete closure cycle — without a line of algebra. It predates Leibniz's formalization of binary arithmetic by roughly four thousand years.

Data and Reproducibility

All algebraic derivations are self-contained and reproducible from the equations presented. The foundational framework, including the full coupling derivation and ghost-freedom proof, is provided in the companion letter [1]. The point-cloud visualization in Figure 1 was generated by evaluating the four tetrahedral half-space inequalities at four million uniformly sampled points; the rendering code is provided as supplementary material. An interactive three-dimensional reconstruction of the dual tetrahedral field chamber (`fig1_dual_tetrahedral_field_chamber.html`) is also included, allowing the reader to rotate, zoom, and inspect the geometry directly. A companion validation script (`run_all_validations.py`) independently verifies every claim in this letter through 42 tests across 16 sections: topology (Sections 1–4), field closure (Sections 5–9), hexagram traversal completeness, cyclotomic structure, N-metric predictions, and the complete derivation chain. All three files are included as supplementary material.

Acknowledgements

The author acknowledges the ghost-free bimetric framework of S. F. Hassan and R. A. Rosen, the tetrad gravity formalism developed by Cartan, Einstein, Weyl, and Ashtekar, and the BBP digit-extraction formula of Bailey, Borwein, and Plouffe.

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